

From Waste to Worth: A Participatory Service System for Furniture Upcycling with Recycled Plastics

Yixuan Li
Shenzhen International Graduate School
Tsinghua University
Shenzhen, China
yixuan-li25@mails.tsinghua.edu.cn

Liu Peixuan
School of Art and Design
Beijing Forestry University
Beijing, China
13503118374@163.com

Jie Zhang*
School of Art and Design
Beijing Forestry University
Beijing, China
jzhang@bjfu.edu.cn

Xin Tong*
The Hong Kong University of Science and Technology
(Guangzhou)
Guangzhou, China
The Hong Kong University of Science and Technology
Hong Kong, Guangdong, China
xint@hkust-gz.edu.cn

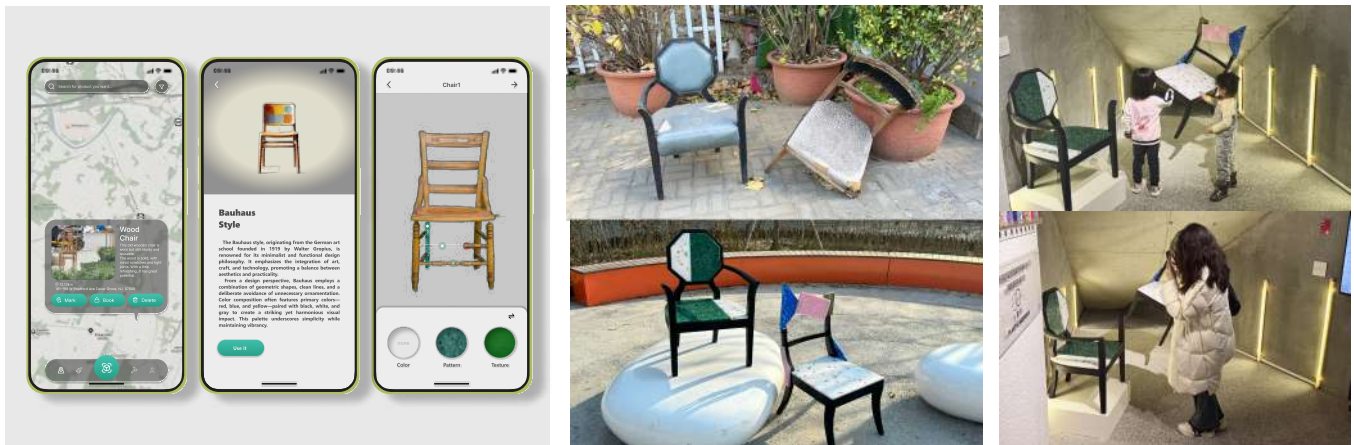


Figure 1: (a) The *Sustainable Furniture* service system. (b) Two chair prototypes developed as design artifacts. (c) Exhibition space used for public feedback collection.

Abstract

Global plastic waste and discarded furniture offer substantial opportunities for resource recovery, yet upcycled products face limited consumer uptake due to stigma toward “waste” materials and doubts about recycled plastic reliability. To mitigate such stigma and reframe perceived value, this study proposes and implements Sustainable Furniture, an interactive service system for upcycling. Grounded in a user-centered approach, the system integrates three mechanisms: material storytelling, traceability visualization, and

creative customization into a participatory workflow spanning furniture selection, material matching, AI-assisted aesthetic customization, and assembly guidance. We report insights from formative user research and iterative usability testing that informed the design. By presenting material provenance, interpretable properties, and curated design templates, the system offers preliminary insights into the potential to reduce stigma toward recycled materials. We present the implemented app and two chair prototypes as design artifacts. Feedback from an exhibition deployment and usability sessions further supports these findings.

Keywords

Upcycling, Recycled Plastic, Discarded Furniture, Participatory Design, Service System

ACM Reference Format:

Yixuan Li, Liu Peixuan, Jie Zhang, and Xin Tong. 2026. From Waste to Worth: A Participatory Service System for Furniture Upcycling with Recycled Plastics. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in*

*Corresponding Author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3798303>

Computing Systems (CHI EA '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798303>

1 Introduction

Global plastic consumption is 450 million metric tons annually, with 350 million metric tons becoming waste each year [9]. While landfilling and incineration remain common, recycling is widely regarded as the most effective option to reduce environmental impacts such as resource depletion and energy use [7], and scaling recycled-plastic use supports SDG 11 and SDG 12 [17]. However, current recycling technologies largely target thermoplastics by leveraging melt processability [6], and often downcycle materials by degrading intrinsic properties—limiting their suitability for high-value applications. Upcycling addresses this limitation by creatively reimagining recycled plastics to add aesthetic and functional value [1, 4], but it requires appropriate product contexts where such value can be realized and sustained across the life cycle [4]. Discarded furniture provides a promising context for high-value upcycling. As furniture replacement accelerates, the volume of discarded furniture continues to grow [8]. Disposal via incineration or landfilling, especially for wooden items, can release substantial CO₂ and toxic gases [17]. Moreover, grassroots practices have emerged: social media accounts in cities such as Barcelona, New York, and Toronto share the locations of curbside items, enabling the reuse of large numbers of furniture pieces. Prior work also highlights upcycling and circular solutions as viable approaches in the furniture industry [10, 14]. Despite this momentum, everyday furniture upcycling remains hard to initiate and sustain due to stigma (e.g., hygiene concerns) [14], visible wear and damage, and uncertainty about the performance and reliability of recycled-plastic components. These barriers motivate the need for structured, design-supported upcycling processes that improve acceptance and enable effective reuse.

In this context, a comprehensive service system and creative upcycling redesign are critical for addressing the stigma and uncertainty that often surround “waste-derived” materials. Research shows that consumers are more likely to purchase remanufactured furniture when it is supported by a structured service process and high-quality refurbishments [5]. Moreover, creative design is key to shifting consumer perceptions, expanding the market for upcycled products, and demonstrating designers’ commitment to environmental sustainability [10]. Building on these insights, we argue that furniture upcycling requires tailored processes that both reduce perceived risk (e.g., unknown material performance) and counter negative associations. Incorporating recycled plastics with transparent material information, alongside participatory design [12], can strengthen user trust and psychological ownership, thereby improving acceptance.

This project develops *Sustainable Furniture*, an app co-designed with recycled-plastic suppliers to support furniture upcycling through a comprehensive service system. It enables users to mark discarded neighborhood furniture, customize redesigns via AI-assisted or manual tools, integrate recycled-plastic components, and receive self-assembly materials. Formative user research and iterative usability testing informed the design. Providing material provenance, interpretable properties, and curated templates, the system reduces



Figure 2: (a)Difficult to locate. (b)Diverse selection. (c)Surface damage or deterioration

stigma toward recycled plastics and improves perceived quality and social acceptability. We present the app and two chair prototypes as design artifacts; exhibition and usability feedback further support these findings.

2 Related Work

Upcycling refers to the process of reusing discarded materials to create products of higher value or quality than the original [15]. In recent years, upcycling has gained attention for its potential to address environmental issues while encouraging creative solutions. Researchers have explored different facets of upcycling across various materials. For example, upcycling waste cotton fabrics into multifunctional e-textiles [16], and exploring electronic waste management practices [11], or using Tree Branches as Architectural Elements [18]. The practice of upcycling can also serve as a tool to raise environmental awareness and inspire creativity. Researcher suggests that upcycling sensitizes individuals to global environmental issues and encourages exploration of innovative solutions [15]. A significant area of upcycling has focused on recycled plastics, with many researchers and designers contributing to this field. For instance, reuse recycled plastic materials in concrete mixes as an environmentally friendly construction material [7]. Choi and Ishii [3] demonstrated the use of wasted thermoplastic bags—commonly found in households—to create patterns and heat-sealed designs for various applications. In addition, the increasing amount of discarded furniture has exacerbated environmental challenges, making it a key area for upcycling research. One creative example is the use of vintage objects, such as clocks, to create playful game controllers [13]. However, such projects often require advanced technical skills or lack an integrated platform to support users throughout the upcycling process. Moreover, there is a growing need for solutions that provide users with a more comprehensive service experience in the upcycling journey [5]. Emphasizes the importance of redesigning discarded materials to meet user needs [2], helping users to engage fully in the process of upcycling and providing creative insights. Such platforms or systems could enable broader participation and make upcycling more accessible.

3 Research and Establishing Requirements

3.1 Survey and Interview

To inform the design, we conducted a 4-hour field study on Xueyuan Road in Beijing to document the quantity, types, and placement of discarded furniture. Wooden chairs, sofas, and bookshelves were most common, typically left near community gathering areas or roadsides. We found that locating items is difficult without precise information about their whereabouts (Fig. 2a), indicating the need

for a map-based system that supports accurate geolocation. We also observed substantial variation in styles and categories (Fig. 2b), suggesting that users would benefit from classified listings and map navigation with style/type filters to find preferred items, thereby improving the user experience. Finally, many items showed damage or deterioration on seat surfaces and backrests (Fig. 2c).

Recycled plastic was selected as a promising material for chair repair and upcycling because it is durable, offers rich color possibilities, and affords creative expression. We then conducted semi-structured interviews with eight participants (Table 1) interested in design or collecting to examine two key issues: (1) how creative redesign with recycled plastics could be supported, and (2) how stigma and perceived risk (e.g., hygiene concerns and perceived quality) shape acceptance of discarded furniture and recycled-plastic components. Participants were introduced to the concept of upcycling and our field-study insights to ground the discussion.

Need for a Comprehensive Service System. All participants (N=8) expressed concerns about two main issues: hygiene (e.g., the furniture found on the roadside may be dirty)(P1, P4) and not knowing how to clean it (P2). However, some participants suggested solutions. For instance, P1 said: *"...If it's from a legitimate second-hand furniture recycling platform or comes with a cleaning kit, I wouldn't see hygiene as a big issue..."* Moreover, it is essential for our product to include modification components to save users' time (P1). As P1 mentioned: *"...It needs to fit my home furnishing style and consider time cost. It would be best if it could be assembled and save me time by handling all the details..."* This highlights the need for a complete service system that includes cleaning and modification tools to simplify the process for users.

Demand for Creative Design Assistance. Participants expressed a strong interest in recycled plastic, finding the material innovative (P5), helpful in reducing plastic waste, and offering various patterns with the potential for infinite recycling (P6). They also expressed a desire for creative design opportunities in recycling old furniture, which aligns with our design approach. Users wanted to redesign furniture to match their personal preferences (P4) or align with their home decor (P1, P2, P8). However, some participants noted difficulty in envisioning their desired design (P8) and suggested that our product can automatically generate design options and offer personalized suggestions (P5). Therefore, it is essential to provide users with customizable creative design assistance to help them realize their vision.

4 Design Process

Insights from our prior research and interviews, we identified two key features: a comprehensive service system and support for creative design. We present our high-fidelity prototype (Fig. 3), along with usability testing and user feedback.

4.1 Design Feature

4.1.1 Comprehensive Service System with Stigma-Reduction Mechanisms. After gaining insights into user needs, we created a storyboard (Fig. 4) to illustrate the end-to-end workflow of *Sustainable Furniture*. The platform enables users to discover discarded furniture nearby, redesign it in-app via generative AI or manual

customization, and order recycled-plastic panels supplied through partner enterprises. The AI component is implemented via OpenAI APIs for (1) design image generation and explanation. Users can also perform controllable, prompt-based editing to refine and optimize the design iteratively. After receiving materials, users assemble the furniture themselves, completing a participatory workflow that integrates selection, redesign, material acquisition, and assembly.

Importantly, the system is designed to mitigate stigma and perceived risk around discarded furniture and recycled plastics. To this end, the interface embeds value-signaling mechanisms at key decision points. It presents provenance cues (e.g., source and processing steps) and interpretable descriptions of material properties and use constraints. It also provides guided preparation and care recommendations (e.g., cleaning and sanding tools) to address hygiene concerns and visible wear. Together, these mechanisms improve perceived quality and social acceptability, supporting participation in furniture upcycling.

4.1.2 Support for Creative Design. To encourage user participation in the design process and allow for customization, users can scan the discarded furniture, and the generative AI will generate various creative design options based on the furniture's appearance (Fig. 3 b, c). Users can also adjust the color, pattern, and texture of the recycled plastic in different designs (Fig. 3 e) to meet their personal preferences. Users can then place an order for recycled plastic panels in the required size and proceed with assembly, contributing to a sustainable impact.

4.2 Usability Testing

We evaluated our High-Fidelity Prototype with six users from Beijing and Guangzhou, each participating in a 40-minute session (Fig. 3). The sessions began with an introduction to the Sustainable Furniture app, followed by tasks where users selected furniture from the map, designed an upcycling concept, and purchased related accessories. Users provided positive feedback on the app's usability, finding our service system clear, intuitive, and visually appealing. They were interested in the idea of involving users in the upcycling process and suggested adding more eco-friendly materials with relevant information to enhance sustainability. The evaluation provides preliminary insights that the platform effectively facilitates user engagement in upcycling and demonstrates scalable potential.

4.3 Practice-Based Analysis: Prototype Design and Exhibition Feedback

To validate our product service workflow and demonstrate creative upcycling with recycled plastics, we developed two chair prototypes (Fig. 5) and gathered public feedback through an exhibition. We sourced two discarded chairs from the streets of Beijing, assessed and cleaned them in our studio, and measured them to determine the need for replacements. Chair A required new front legs, a backrest, and a seat, while Chair B required a new backrest and seat. We iteratively redesigned concepts with generative AI and refined the selected directions in Procreate, using high-quality visual outcomes as value signals to counter associations with "waste". The final designs feature modern curves with bold color contrasts for Chair A and a classical silhouette with white-green recycled plastic panels

Table 1: Participant Information

ID	Gender	Age	Education	Field of Study
P1	F	23	Undergraduate	Visual Communication Design
P2	F	22	Undergraduate	Social Work
P3	F	50	Undergraduate	Financial
P4	M	22	Undergraduate	Artificial Intelligence
P5	F	22	Undergraduate	Product Design
P6	M	47	Bachelor's Degree	Clinical Medicine
P7	M	26	Postgraduate Degree	Cross-cultural Education
P8	F	22	Undergraduate	Visual Communication Design

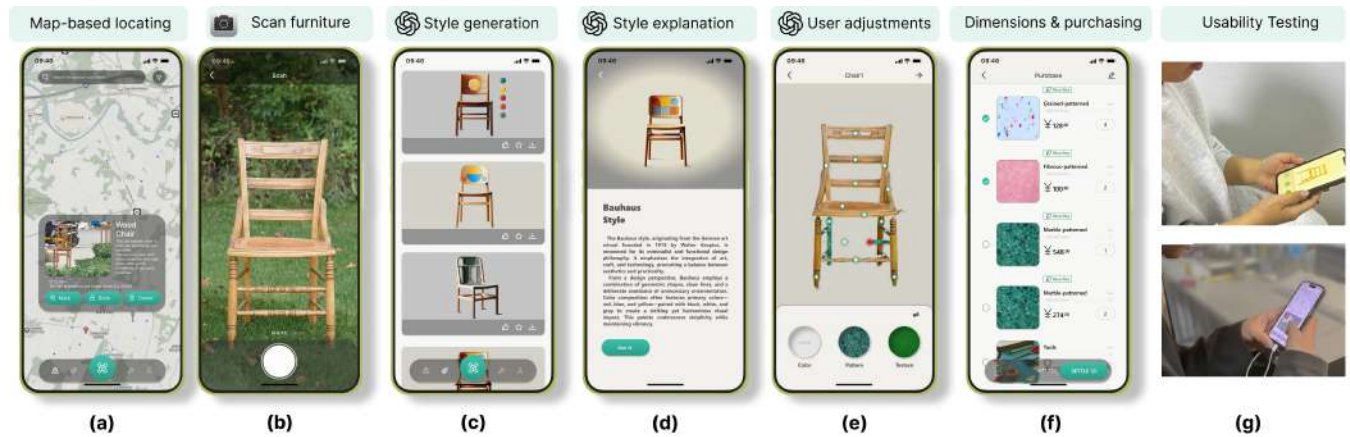


Figure 3: Hi-fi Prototype: (a) Users locate discarded furniture on a map. (b) After pickup, users scan the furniture. (c) AI generates recycled-plastic style options. (d) The system explains design options. (e) Users adjust colors and patterns. (f) Users enter dimensions and order materials and prep tools (e.g., cleaning and sanding).

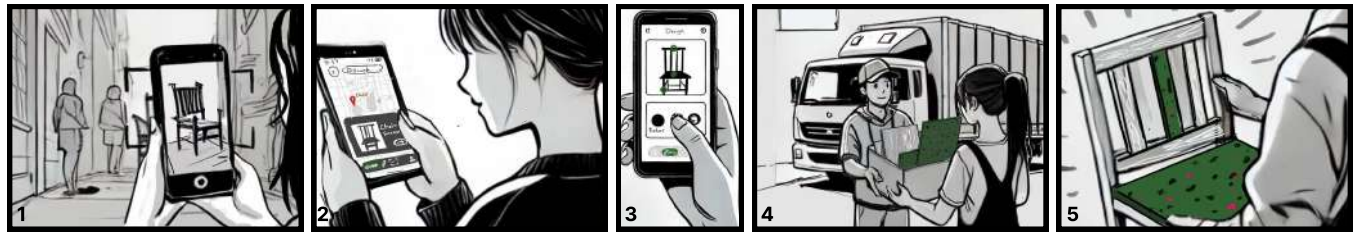


Figure 4: Storyboard: (a) User A spots discarded furniture and uploads it to the app. (b) User B likes the furniture and retrieves it. (c) User B designs the furniture style in the app. (d) User B orders recycled plastic. (e) User B upcycles the furniture.

for Chair B. We then ordered recycled plastic panels from our suppliers, which were cut and installed to specification.

We exhibited the prototypes at the Beijing Wind Hart Center (Fig. 1). Approximately 100 visitors attended each week, and we conducted two hours of observation and interviews on one day. Many young adults and children showed strong interest, and visitors highlighted the aesthetic contrast between wood and recycled plastic, as well as the works' artistic appeal and storytelling. One attendee noted: "...I think upcycling furniture on your own is highly engaging

and socially meaningful...the contrast between the recycled plastic material and the original wooden components is thought-provoking and impactful..." These findings offer preliminary insights into the potential for reducing stigma toward recycled materials. By emphasizing the aesthetic value of reused components, the design improves the perceived acceptability of discarded furniture, suggesting a feasible way to shift social perceptions.



Figure 5: Upcycling Prototype Design Process: (1) Pick up Discarded Furniture. (2) Furniture Cleaning. (3) Measure Repair Areas. (4) Design. (5) Develop Prototypes. (6) Product Display

5 Conclusion and Future Work

This paper presented *Sustainable Furniture*, an interactive service system for furniture upcycling that integrates recycled plastics into user-customizable designs. We highlight two contributions: (1) interaction support for creative redesign, and (2) stigma reduction through value signaling (e.g., provenance cues, interpretable material properties and constraints, curated templates, and preparation guidance) to improve perceived quality and social acceptability. We demonstrate feasibility through the implemented app, two chair prototypes, and feedback from usability sessions and an exhibition deployment. Future work will involve longer-term, in-the-wild deployments to understand what motivates users to upload and retrieve discarded furniture, how stigma evolves over time, and which value signaling mechanisms best support sustained engagement.

Acknowledgments

We thank Jia Li (Sophia) and Beijing Plastic Response Information Technology Co., Ltd. for their support in this work.

References

- [1] Spyros Bofylatos. 2022. Upcycling systems design, developing a methodology through design. *Sustainability* 14, 2 (2022), 600.
- [2] Mengyu Cao. 2017. Lunhui: Upcycled Creations. In *Proceedings of the 2017 ACM SIGCHI Conference on Creativity and Cognition*. 269–271.
- [3] Kyung Yun Choi and Hiroshi Ishii. 2021. Therms-up!: Diy inflatables and interactive materials by upcycling wasted thermoplastic bags. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–8.
- [4] Louis Diep. 2024. Reimagining waste: Upcycling Plastic Waste in Product Design.
- [5] Giliam Dokter, Liane Thuvander, and Ulrike Rahe. 2021. How circular is current design practice? Investigating perspectives across industrial design and architecture in the transition towards a circular economy. *Sustainable Production and Consumption* 26 (2021), 692–708.
- [6] Jeannette M Garcia and Megan L Robertson. 2017. The future of plastics recycling. *Science* 358, 6365 (2017), 870–872.
- [7] Lei Gu and Togay Ozbakkaloglu. 2016. Use of recycled plastics in concrete: A critical review. *Waste Management* 51 (2016), 19–42. doi:10.1016/j.wasman.2016.03.005
- [8] Tianhui Huang, Tianming Ke, Yuyang Lin, Fangya Shi, and Xinyu Zhang. 2023. Green Home: A Reflection on the Relationship between Human and Furniture. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 584, 5 pages. doi:10.1145/3544549.3583833
- [9] Sanjay Kumar Joshi and Akshita Vashishth. 2024. Plastonomics: Impact of Plastic on Ecosystem and the World Economy. In *Advanced Strategies for Biodegradation of Plastic Polymers*. Springer, 405–420.
- [10] Sua Lee. 2019. *Sustainable Design Approaches Using Waste Furniture Materials for Design Practitioners*. Ph. D. Dissertation. Buckinghamshire New University (Awarded by Coventry University).
- [11] Jasmine Lu and Pedro Lopes. 2024. Unmaking electronic waste. *ACM Transactions on Computer-Human Interaction* 31, 6 (2024), 1–30.
- [12] Michael J Muller and Sarah Kuhn. 1993. Participatory design. *Commun. ACM* 36, 6 (1993), 24–28.
- [13] Jan Pokorny, Magdalena Kejstova, Vit Rusnak, and Simone Kriglstein. 2023. From bin to playin': Give vintage objects a new purpose as game controllers. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [14] Anh Tran-Nevalainen. 2024. Finnish consumers in the furniture sector towards the circular economy—a consumer culture theory approach. (2024).
- [15] Anne Weibert, Marios Mouratidis, Renad Khateb, Sarah Rüller, Miriam Hosak, Shpresa Potka, Konstantin Aal, and Volker Wulf. 2017. Creating environmental awareness with upcycling making activities: A study of children in Germany and Palestine. In *Proceedings of the 2017 conference on interaction design and children*. 286–291.
- [16] Irmandy Wicaksono, Aditi Maheshwari, Don Derek Haddad, Joseph Paradiso, and Andreea Danielescu. 2024. Design and Fabrication of Multifunctional E-Textiles by Upcycling Waste Cotton Fabrics through Carbonization. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 2 (2024), 1–31.
- [17] Xianqing Xiong, Xinyi Yue, Weihang Dong, and Zhenzhong Xu. 2022. Current status and system construction of used-furniture recycling in China. *Environmental Science and Pollution Research* 29, 55 (2022), 82729–82739.
- [18] Hironori Yoshida, Maria Larsson, and Takeo Igarashi. 2019. Upcycling tree branches as architectural elements through collaborative design and fabrication. In *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 589–593.